

$$\begin{aligned}
 P_2 &= \sum_{a=1}^N \vec{p}_a \cdot \frac{\partial \vec{r}_a}{\partial \dot{q}_2} \\
 &= \sum_{a=1}^N \vec{p}_a \cdot (\hat{n} \times \vec{r}_a) \\
 &= \sum_{a=1}^N (\vec{r}_a \times \vec{p}_a) \cdot \hat{n} \\
 &= \hat{n} \cdot \left(\sum_{a=1}^N \vec{L}_a \right) = \hat{n} \cdot \vec{L}
 \end{aligned}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_2} \right) = \frac{\partial L}{\partial q_2}$$

$$\begin{aligned}
 \frac{dP_2}{dt} &= \frac{\partial L}{\partial q_2} = Q_2 = \sum_{a=1}^N \vec{F}_a \cdot \frac{\partial \vec{r}_a}{\partial \dot{q}_2} \\
 &= \sum_{a=1}^N \vec{F}_a \cdot (\hat{n} \times \vec{r}_a) \\
 &= \sum_{a=1}^N (\vec{r}_a \times \vec{F}_a) \cdot \hat{n} = \hat{n} \cdot \sum_{a=1}^N \vec{N}_a \\
 &= \vec{N} \cdot \hat{n}
 \end{aligned}$$

$$\frac{\partial U}{\partial \dot{q}_2} = 0 \Rightarrow P_2 = \frac{\partial L}{\partial \dot{q}_2} = \frac{\partial L}{\partial \dot{q}_2} = \vec{L} \cdot \hat{n}$$

$$\boxed{\frac{\partial \vec{r}_a}{\partial \dot{q}_2} = \hat{n} \times \vec{r}_a}$$

$$\begin{aligned}
 \frac{dP_2}{dt} &= \frac{\partial L}{\partial q_2} = \vec{N} \cdot \hat{n} \\
 \frac{d}{dt} (\vec{L} \cdot \hat{n}) &= \frac{\partial L}{\partial q_2} = \vec{N} \cdot \hat{n}
 \end{aligned}$$

Tenemos

$$\vec{r} = x \hat{x}$$

$$L = \frac{m}{2} \dot{x}^2 ; \quad \frac{\partial L}{\partial \dot{x}} = m \dot{x} = p_x$$

$$p_x \equiv \frac{\partial L}{\partial \dot{x}}$$

$$p_i \equiv \frac{\partial L}{\partial \dot{q}_i}$$

$$\frac{\partial L}{\partial \dot{x}} = m \dot{x} ; \quad \dot{x} \frac{\partial L}{\partial \dot{x}} = m \dot{x}^2 = 2T$$

$$\dot{x} \frac{\partial L}{\partial \dot{x}} = 2L$$

$$\Rightarrow \dot{x} \frac{\partial L}{\partial \dot{x}} - L = L$$

Función h

$$h = \sum_{i=1}^n \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L$$

→ Transformada de Legendre

$$L = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} m \omega^2 x^2$$

$$h = \dot{x} \frac{\partial L}{\partial \dot{x}} - L$$

$$= \dot{x} m \dot{x} - \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2$$

$$h = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2 = T + U = E, = \vec{N} \cdot \hat{n}$$

Para partícula libre

$$P.L. \} h = E = \frac{1}{2} m \dot{x}^2$$

$$\frac{dh}{dt} = \dot{E} = \frac{1}{2} m 2 \dot{x} \ddot{x} = p \ddot{x} = 0$$

$$\frac{dh}{dt} = \frac{d}{dt} \left[\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2 \right] = \frac{1}{2} m 2 \dot{x} \ddot{x} + \frac{1}{2} m \omega^2 2 x \dot{x}$$

Tenemos

Factorizando ...

$$= m \dot{x} \left[\ddot{x} + \omega^2 x \right] = 0$$

Ec. de movimiento para
un oscilador armónico no amortiguado ni forzado

$$\ddot{x} + \omega^2 x = 0$$

$$\ddot{x} = 0$$

Ec. de movimiento para
partícula libre.

Tenemos

$$\frac{dh}{dt} = \frac{d}{dt} \sum_{i=1}^n \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - \frac{dL}{dt}$$

$$= \sum_{i=1}^n \frac{d}{dt} \left(\dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) - \frac{dL}{dt}$$

Distribuyendo la derivada.

$$\sum_{i=1}^n \left[\underbrace{\frac{d}{dt}(\dot{q}_i)}_{\ddot{q}_i} \cdot \frac{\partial L}{\partial \dot{q}_i} + \dot{q}_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right] - \frac{dL}{dt}$$

Tenemos ...

$$= \sum_{i=1}^n \left[\ddot{q}_i \frac{\partial L}{\partial \dot{q}_i} + \dot{q}_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right] - \sum_{i=1}^n \left[\frac{\partial L}{\partial \dot{q}_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \right] - \frac{\partial L}{\partial t}$$

Dependencia de la Lagrangiana

$(q, \dot{q}, t)$